

CLOUDCORE SOLUTIONS

Advanced Cloud & Data Analytics

Response to Request for Proposal

Migration & Modernisation of Manga's Data Architecture to the Cloud

The Manga DataHub — a cloud-native AWS data lakehouse

Submitted to: Manga S.L. — Barcelona, Spain

IE University · MBD-EN2025 · Data Analytics in the Cloud · Group Work Option B

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1. Executive Summary

Manga stands at an inflection point. With annual revenues of **€280 million** across 100+ stores worldwide, the brand has a strong foundation — yet growth of only **3.5%** trails the market, online sales grew just **1.2% year-over-year**, and the company spends roughly **twice as much as competitors** for half the analytical output. The root cause is architectural: independent ETL pipelines triggered by daily cron jobs, aggregated into Excel files shared once per day, cannot power the real-time, personalised experiences modern retail demands.

CloudCore Solutions proposes the **Manga DataHub** — a cloud-native AWS data lakehouse that replaces this legacy stack with a unified, scalable, secure platform built around three strategic pillars:

- **Unify** — centralise every data source into a single governed platform, replacing siloed pipelines and manual Excel reporting.
- **Accelerate** — cut time-to-insight from 24 hours to sub-second, enabling live inventory, sales dashboards, and fraud detection.
- **Enable** — deliver the ML features on Manga's roadmap (recommendations, dynamic pricing, demand forecasting, virtual assistant) impossible today.

The solution is built entirely on AWS using open-source tooling (Apache Airflow, Apache Spark, open Parquet) to guarantee portability and eliminate vendor lock-in — directly addressing the CTO's concerns. Three isolated environments (Dev, Pre-Prod, Prod) are provisioned from identical Infrastructure-as-Code templates, with anonymised production samples in non-production. Security is designed in from day one: column-level access control, automated PII scanning, KMS encryption, and full GDPR audit trails.

Expected outcomes within 12 months: a **40–60% reduction in infrastructure cost** versus the colocation model, real-time reporting replacing the daily batch cycle, and the first ML use cases live within six months of kickoff.

Current State vs. Manga DataHub

Dimension	Current State	With Manga DataHub
Reporting lag	24 hours (daily batch)	Sub-second (real-time streaming)
Infrastructure cost	≈2× competitor average	Projected 40–60% reduction
BI access	Manual Excel files	Self-service dashboards
ML capabilities	None	Forecasting, recommendations, pricing
Data environments	Single (production only)	Dev / Pre-Prod / Prod (isolated)
GDPR compliance	Unclear	Automated PII scanning & masking

2. High-Level Architecture

The Manga DataHub is organised into five core layers with three cross-cutting concerns and a clear separation between batch and streaming paths. This conceptual view is technology-agnostic; each decision is grounded in foundational cloud-analytics principles.

Architectural Layers

- **Data Sources** — POS/ERP transactional data, web & mobile event streams, CRM/loyalty records, and external APIs spanning structured, semi-structured, and unstructured modalities.
- **Ingestion** — a streaming path for real-time events and a batch path for scheduled loads, plus a decoupled API-connector layer so new sources are added without touching existing pipelines.
- **Storage — the Lakehouse** — a three-zone medallion model: Raw (immutable system of record), Curated (cleaned, validated, PII-masked), and Serving (pre-aggregated, query-optimised).
- **Processing** — distributed transformation, DAG-based orchestration replacing all cron jobs, automated post-pipeline data-quality checks, and a central catalog for lineage and discovery.
- **Consumption** — self-service BI dashboards, a managed ML environment with a feature store, and secure versioned APIs for partners and AI agents.

Cross-Cutting Concerns

Security & Compliance (encryption, RBAC, PII detection, GDPR audit, threat detection), Infrastructure-as-Code & Automation (version-controlled, three identical environments), and Cost & Sustainability (auto-scaling, lifecycle archiving, per-service carbon tracking) apply across every layer — not as optional add-ons.

Design Principles

Principle	Application in Manga DataHub
Separation of Concerns	Each layer has a single responsibility; storage is decoupled from compute.
Modularity	New sources, jobs, and models are isolated modules — a failure in one does not cascade.
Elasticity	Compute scales automatically; a Fashion-Week spike needs no manual intervention.
Scalability	Handles current volumes and the 10× growth from international expansion.
Open Standards	Parquet, Airflow, Spark — open formats and tooling at every layer; no lock-in.

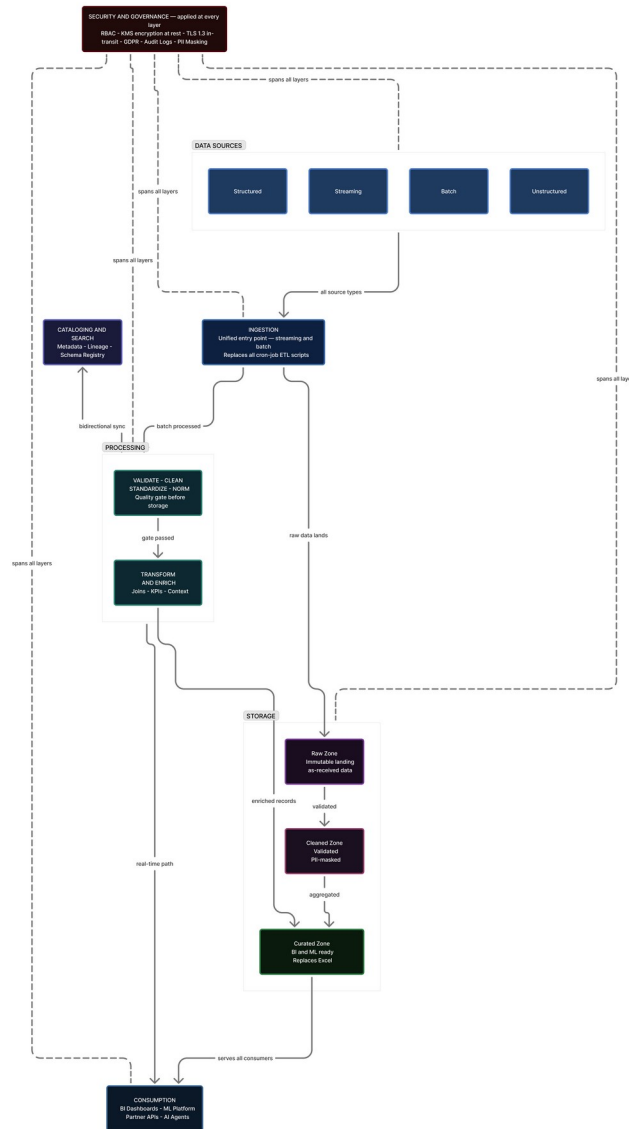


Figure 1 — High-Level Architecture (technology-agnostic).

3. Low-Level Architecture

The conceptual layers map to specific AWS services, each chosen for performance, operational reliability, cost model, and requirement fit.

Ingestion

- **Amazon Kinesis Data Streams + Firehose** — sub-second ingestion of POS and web events; Firehose delivers micro-batches to S3 in Parquet (R1, R3).
- **AWS Glue ETL** — managed Spark replacing hand-coded batch scripts; job bookmarks prevent duplicate processing (R1, R3).
- **AWS Lambda + EventBridge** — serverless connectors for third-party APIs; new sources are new functions, no pipeline changes (R7).
- **AWS DMS** — one-time historical migration plus CDC replication for zero-loss cutover (R3, R4).

Storage

- **Amazon S3 — three-zone lakehouse** — Raw, Curated (PII-masked Parquet), Serving; Intelligent-Tiering cuts cold-storage cost; open format means zero lock-in (R1, R6).
- **Amazon Redshift** — columnar warehouse; Spectrum queries S3 directly; RA3 decouples compute and storage (R1, R6).
- **Amazon DynamoDB** — sub-10ms lookups for ML inference and the virtual shopping assistant (R1, R7).

Processing & Governance

- **Glue Data Catalog, Amazon MWAA** (Airflow orchestration replacing cron), **Glue Data Quality, Lake Formation** (column/row RBAC, GDPR), and **Amazon Macie** (ML-powered PII scanning).

Consumption & Cross-Cutting

- **Consumption:** QuickSight (BI), SageMaker (ML), Comprehend (NLP sentiment), Athena (serverless SQL), API Gateway + Lambda (versioned external APIs).
- **Cross-cutting:** IAM + KMS (TLS 1.3, encryption at rest), Terraform/CDK (IaC), CloudWatch + SNS (monitoring), AWS Organizations (isolated accounts), Carbon Footprint Tool (R8).

4. Project Implementation

Requirements Traceability (R1–R8)

Req.	Requirement	How it is addressed
R1	Unified multi-modal platform	S3 lakehouse + Redshift + DynamoDB store all data modalities; Kinesis (streaming) and Glue (batch); a single Glue Catalog provides unified access.
R2	Multi-environment setup	AWS Organizations provisions Dev / Pre-Prod / Prod from identical Terraform templates; Glue masking + Lake Formation replicate anonymised samples to non-prod.
R3	Automation & IaC	MWAA replaces cron with monitored DAGs; Terraform/CDK automates infrastructure; Glue bookmarks and S3 lifecycle policies automate the data lifecycle.
R4	Security & high availability	IAM least-privilege, KMS encryption, Lake Formation RBAC, Macie PII scanning, CloudTrail audit, multi-AZ; target 99.9% uptime.
R5	Data quality	Glue Data Quality rules run post-pipeline; failures trigger CloudWatch/SNS alerts before bad data reaches consumers.
R6	Cost optimisation	Pay-per-use vs. fixed colocation; Intelligent-Tiering, Spot Glue, Redshift RIs, QuickSight per-session — projected 40–60% reduction (see figure).
R7	Extensibility & interoperability	Decoupled Lambda connectors, API Gateway, open Parquet/Airflow formats, and versioned APIs for backward compatibility.
R8	Sustainability	EU renewable-heavy regions; serverless and auto-scaling eliminate idle resources; Carbon Footprint Tool quantifies improvement.

Cost Impact

Replacing the fixed colocation model with AWS pay-per-use is projected to reduce annual infrastructure cost by roughly **57%** — from about **€1.30M to €0.56M per year**, saving on the order of **€0.74M annually**. The figure below breaks the saving down by category.

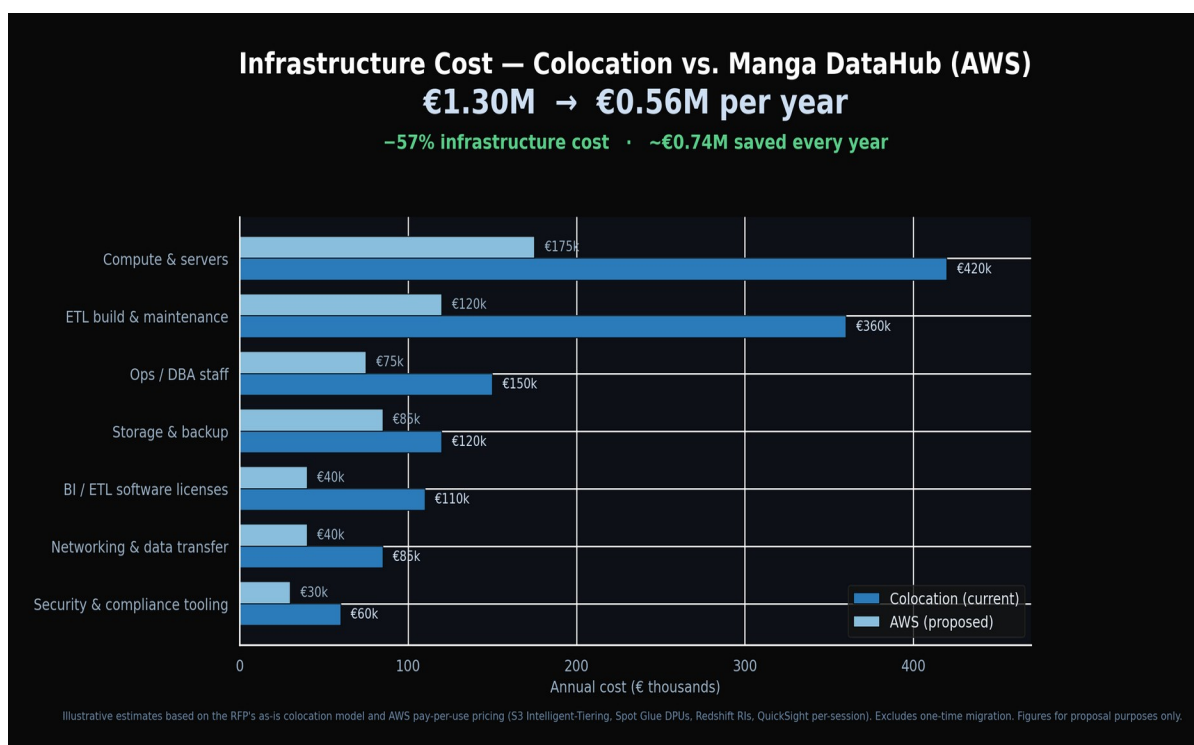


Figure 3 — Projected annual infrastructure cost: colocation vs. Manga DataHub (illustrative).

Methodology & Timeline

CloudCore applies a DataOps methodology — iterative, test-driven, with CI for both pipelines and infrastructure code. Data contracts are validated between layers, and a quality gate blocks promotion from Curated to Serving until thresholds are met. Delivery runs across three phases over nine months.

Phase	Months	Key milestone
1 · Foundation	1–3	AWS account structure, Terraform baseline, Dev live, DMS migration, Catalog + MWAA, first batch pipelines.
2 · Build & Validate	4–6	Pre-Prod live, Kinesis streaming, Glue DQ + Lake Formation, QuickSight replacing Excel, first ML model.
3 · Prod Cutover & ML	7–9	Production live, colocation decommissioned, recommendations + dynamic pricing live, security audit, handover.

Key Risks & Mitigations

- **Migration data loss** — DMS CDC runs in parallel for 4 weeks; automated row-count and checksum reconciliation before cutover.
- **Vendor lock-in (CTO)** — open-source tooling at every layer; architecture documented as cloud-portable.
- **GDPR non-compliance** — Lake Formation + Macie in Phase 1; a DPIA completed before any personal data enters the platform.

5. Sample Use Cases

Three end-to-end use cases demonstrate the architecture using Manga's actual data sources, each spanning ingestion through business outcome.

5.1 Demand Forecasting

Glue joins sales history with external factors (weather, holidays, events, campaigns); features land in a SageMaker Feature Store; DeepAR retrains weekly via an MWAA DAG, producing 7/14/28-day forecasts per SKU per store that drive automated replenishment. **Projected impact: –30% stockouts, –20% overstock write-offs.**

5.2 Product Recommendation Engine

Real-time purchase events stream through Kinesis into a customer-product interaction matrix; SageMaker trains a collaborative-filtering model; results are filtered against live inventory (DynamoDB) and served via API Gateway at ≤5ms. **Projected impact: +12–18% average order value.**

5.3 Sentiment-Driven Merchandising

Review events stream via Kinesis; Amazon Comprehend scores sentiment; Glue aggregates by product, store, and rolling 7-day window; QuickSight surfaces declining products and SNS alerts the buying team. **Projected impact: 2–3 weeks' earlier warning on quality issues.**

Two further use cases extend the same platform: a **dynamic pricing engine** (+3–7% gross margin) and **return-rate optimisation** (–10–15% return-processing cost).

6. Future Improvements

Short-Term (Months 10–18)

- Real-time, session-level personalisation across web and in-store touchpoints (reuses Kinesis + DynamoDB).
- Supply-chain smart labels — RFID/smart-label streams for item-level inventory.
- Advanced data quality — Great Expectations-style suites with statistical distribution checks.

Medium-Term (Months 18–30)

- Virtual shopping assistant on Amazon Bedrock (RAG over catalog, inventory, purchase history).
- External data monetisation — anonymised retail-trend data via an API marketplace.
- Data mesh — domain teams own and publish data products through the shared catalog.

Long-Term (30+ Months)

- AI-agent ecosystem via Bedrock Agents; federated ML across partners without sharing raw data; board-reportable carbon-neutrality roadmap.

Appendix

A. References

- Precedence Research — Retail Analytics Market Size, Share & Trends, 2024–2034.
- NTT Data — Mango Case Study: AWS Cloud Migration.
- AWS — Kinesis, Glue, SageMaker, Lake Formation developer documentation.
- Manga DataHub interactive platform (CloudCore Solutions) — live on Vercel.

B. Generative AI Acknowledgement

This document was produced with the assistance of generative AI tools in accordance with IE University MBD-EN2025 course guidelines. AI assistance supported drafting and refining written sections, validating AWS service recommendations, and producing supporting visualisations. All architectural decisions, analysis, and rationale were reviewed, validated, and approved by the team, which retains final responsibility for all content.

C. Team Contributions

Member	Primary responsibilities
Andrea Alarcón Valles	Executive Summary, document coordination, stakeholder analysis, final editing.
Mateus Carneiro	High-Level Architecture, conceptual design, architecture diagrams.
Tina Jannasch	Low-Level Architecture, AWS service selection and cost rationale.
Ricardo Liévano Pedroza	Project Implementation, R1–R8 mapping, risk register, timeline.
Luka Tcheishvili	Data-source analysis, sample use cases, pipeline design, interactive web app.
Nicklas Urban	Future Improvements, GitHub repository, deployment, presentation lead.

D. Operating Evidence Appendix

The following operating evidence turns the R1-R8 requirements into measurable delivery controls. Cost and emissions figures are planning assumptions to validate during discovery; the monitoring, retention, quality, and recovery controls are proposed operating targets for production.

R6 Cost Model Assumptions

Cost area	Current colocation assumption	AWS target assumption	Saving lever
Compute and ETL	Always-on partner servers sized for peak nightly jobs.	Glue billed per DPU-hour, Spot for non-critical jobs, and small MWAA schedulers.	Pay only when pipelines run.
Storage and backup	SAN/NAS plus backup contract sized for seven-year growth.	S3 Raw, Curated, and Serving zones with Intelligent-Tiering and Glacier lifecycle.	Hot data stays fast; cold data becomes cheap.
BI and serving	Excel distribution plus manual engineering support.	QuickSight pay-per-session, Athena per-query, and Redshift RA3 for steady BI.	Usage-based analytics.
Operations	Partner-operated colocation and fixed support contract.	Managed AWS services, CloudWatch/SNS automation, and IaC repeatability.	Fewer manual runbooks and escalations.
Model result	About EUR 1.30M per year baseline.	About EUR 0.56M per year AWS target.	40-60% range; 57% base-case reduction validated monthly with Cost Explorer and CUR.

R8 Sustainability Metrics

Metric	Baseline method	Target / control	Cadence
Operational carbon index	Normalize current colocation to 100 using invoices, server count, provider PUE, and energy mix.	Target index 45-60 through AWS Customer Carbon Footprint Tool reporting.	Monthly
Idle compute ratio	Always-on servers sized for peak batch windows.	Less than 20% idle compute through serverless, autoscaling, and scheduled jobs.	Monthly
Storage efficiency	All historical data held in high-cost tiers.	90-day S3 IA and 180/365-day Glacier transitions.	Monthly
Region policy	Provider-dependent energy mix.	Prefer EU Ireland / Frankfurt; enforce region choices through IaC variables and SCP guardrails.	Every release
Reporting	Limited provider transparency.	Dashboard by service, environment, kgCO2e, TB-month, DPU-hour, and EUR spent.	Monthly ESG pack

R4 HA / DR Targets

Workload	HA / DR pattern	RPO	RTO
S3 lakehouse	Versioning, Object Lock on Raw, lifecycle policies, optional cross-region replication for Raw and Curated zones.	15 min	4 h
Kinesis / Firehose	Managed Multi-AZ stream, retries, failed-record S3 prefix, and replay from Raw zone.	<5 min	<1 h
Glue / MWAA	DAGs and jobs in Git, Terraform redeploy, Glue bookmarks, retries, and backfills.	Last successful checkpoint	4 h
Redshift / Athena / QuickSight	Automated Redshift snapshots, Spectrum over S3, and dashboards that can repoint to standby warehouse.	15 min-24 h by data mart	4-8 h
DynamoDB / APIs	Point-in-time recovery, autoscaling, and optional global tables for customer-facing APIs.	<5 min	<1 h

R5 Data Quality Rule Examples

Dataset	Concrete data quality controls
sales_sample.csv	purchase_price > 0; customer_id completeness > 99%; product_id referential integrity to inventory.
customers_sample.csv	customer_id complete and unique; age between 13 and 100; PII detection before non-prod replication.
inventory_sample.csv	stock >= 0; price > 0; product_description completeness > 95%.
customer_reviews_sample.csv	Review text length between 5 and 2,000 characters; rating between 1 and 5; product/customer IDs complete.
external_factors_sample.csv	date complete and unique; fuel price positive; campaign IDs conform to agreed pattern when present.
shipping_sample.csv	order ID complete; actual shipping cost non-negative; returned flag limited to true/false values.

Data Lifecycle and Retention Matrix

Dataset	Sensitivity	Raw zone	Curated / Serving	Archive / deletion policy
Sales	Tokenized customer IDs; financial records	7 years	5 years curated; 3 years serving aggregates	IA after 90 days, Glacier after 365 days; preserve finance aggregates.
Customers	Direct PII	Active relationship plus 24 months	Masked/tokenized in curated; non-prod receives anonymized samples only	Right-to-erasure workflow tokenizes or deletes PII within 30 days, subject to legal hold.
Inventory	No PII	5 years	3 years curated and serving	IA after 180 days; expire obsolete product snapshots after retention.
Customer reviews	PII risk in text/images	3 years	3 years curated sentiment/features	PII scan on landing; remove or anonymize review content on customer request.
External factors	No PII	5 years	5 years curated	IA after 180 days; retain for forecast reproducibility.
Shipping	Address PII and cost records	6 years	3 years curated with addresses masked	IA after 90 days, Glacier after 365 days; delete/mask addresses under privacy workflow.

Appendix: Detailed Cost Analysis

Source-backed TCO model and calculation backup for the Manga DataHub RFP response.

Date prepared: 2026-06-17

This document explains how the project cost estimate was built, which AWS public prices were used, and how the headline claim is calculated:

Current colocation run-rate: EUR 1,300,000 / year
Target AWS run-rate: EUR 560,000 / year
Estimated saving: EUR 740,000 / year
Estimated reduction: $740,000 / 1,300,000 = 56.9\%$, rounded to 57%

1. Important Positioning

The cost model is a presentation-ready directional TCO estimate, not an audited quote. The RFP case gives us two hard inputs:

Input from case	How it is used
Manga has EUR 280M annual revenue	Used only to sanity-check scale. The current cost model equals about 0.46% of revenue.
Manga has about 2x infrastructure cost vs competitors	Used to derive a current baseline from a competitor-normal operating assumption.

The case does not provide Manga's actual colocation invoice, exact data volumes, current BI license spend, or current support contract. Because of that, the absolute EUR values are explicit assumptions. The percentage logic is still useful because it compares fixed, peak-sized infrastructure with managed, usage-based cloud services.

If the professor asks "where did the EUR 1.30M come from?", the honest answer is:

The RFP states that Manga pays about 2x competitors for infrastructure.
We assumed a competitor-normal analytics infrastructure cost of about EUR 650k/year.
Therefore Manga current baseline = $\text{EUR } 650\text{k} * 2 = \text{EUR } 1.30\text{M/year}$.

This should be replaced with Manga's real invoices during discovery.

2. AWS Price Sources Used

All AWS prices below are public AWS pricing references accessed on 2026-06-17. AWS states that prices vary by Region, so the final client estimate should be rebuilt in AWS Pricing Calculator for the chosen EU Region, most likely eu-west-1 Ireland or eu-central-1 Frankfurt.

Service	Public unit price used in this model	Source
AWS Glue ETL, Crawlers, Data Quality	USD 0.44 per DPU-hour	https://aws.amazon.com/glue/pricing/
AWS Glue Data Catalog	First 1M metadata objects free; USD 1.00 per 100k objects over 1M per month	https://aws.amazon.com/glue/pricing/
Amazon Kinesis Data Streams, provisioned	USD 0.015 per shard-hour; USD 0.014 per 1M PUT payload units in AWS example	https://aws.amazon.com/kinesis/data-streams/pricing/
Amazon Kinesis Data Streams, on-demand standard example	USD 0.08 per GB data-in; USD 0.040 per GB data-out in AWS example	https://aws.amazon.com/kinesis/data-streams/pricing/
Amazon Data Firehose	USD 0.029 per GB ingested for first 500 TB/month; USD 0.018 per GB for JSON to Parquet/ORC conversion; USD 0.020 per GB dynamic partitioning	https://aws.amazon.com/firehose/pricing/

Service	Public unit price used in this model	Source
AWS Lambda	USD 0.20 per 1M requests; USD 0.0000166667 per GB-second; first 1M requests and 400k GB-seconds free per month	https://aws.amazon.com/lambda/pricing/
Amazon S3 Standard	USD 0.023 per GB-month planning rate	https://aws.amazon.com/s3/pricing/
Amazon S3 Standard-IA	USD 0.0125 per GB-month planning rate	https://aws.amazon.com/s3/pricing/
Amazon S3 Glacier Flexible Retrieval	USD 0.0036 per GB-month planning rate	https://aws.amazon.com/s3/pricing/
S3 lifecycle transition	USD 0.01 per 1,000 transition requests from S3 Standard to Standard-IA, per AWS example	https://aws.amazon.com/s3/pricing/
Amazon Athena SQL	USD 5 per TB scanned; AWS example shows 3 TB = USD 15 and Parquet/column pruning reducing scan cost	https://aws.amazon.com/athena/pricing/
Amazon Athena capacity reservation	USD 0.30 per DPU-hour in AWS example	https://aws.amazon.com/athena/pricing/
Amazon Redshift provisioned/serverless	Provisioned starts at USD 0.543/hour; Serverless begins at USD 1.50/hour	https://aws.amazon.com/redshift/pricing/
Amazon Redshift managed storage	USD 0.024 per GB-month in AWS example	https://aws.amazon.com/redshift/pricing/
Amazon QuickSight Author	USD 24 per user/month	https://aws.amazon.com/quicksight/pricing/
Amazon QuickSight Reader	USD 3 per user/month	https://aws.amazon.com/quicksight/pricing/
Amazon QuickSight SPICE	USD 0.38 per GB/month; 10 GB included with each Author	https://aws.amazon.com/quicksight/pricing/
Amazon MWAA	AWS examples show about USD 449/month for a small environment and about USD 1,047/month for a large environment	https://aws.amazon.com/managed-workflows-for-apache-airflow/pricing/
AWS DMS	Serverless/on-demand migration capacity is pay-per-hour; AWS Pricing Calculator recommended for final migration sizing	https://aws.amazon.com/dms/pricing/

Planning FX assumption for presentation tables:

1 USD = 0.92 EUR

This FX rate is a planning convention only. Replace it with the current EUR/USD rate before submission if the team wants exact currency conversion.

3. Current Colocation Baseline

The baseline is not from an invoice. It is derived from the RFP fact that Manga pays about 2x competitors for infrastructure.

3.1 Baseline Formula

```

Annual revenue = EUR 280,000,000
Assumed competitor-normal analytics infrastructure intensity = 0.23% of revenue

Competitor-normal annual infrastructure cost
= 280,000,000 * 0.0023
= EUR 644,000
Rounded planning baseline = EUR 650,000

Manga current infrastructure cost
= competitor-normal baseline * 2
= 650,000 * 2
= EUR 1,300,000 per year

```

3.2 Baseline Cost Breakdown

Current cost area	Annual cost	Why this is plausible for current state
Compute and ETL servers	EUR 420,000	Always-on partner/colocation servers sized for peak nightly batch jobs, cron scripts, and manual reprocessing.
Storage and backup	EUR 280,000	SAN/NAS, backup retention, duplicate extracts, and historical data kept in expensive tiers.
BI and reporting	EUR 240,000	Excel distribution, manual report generation, licensing/support overhead, and engineering time for every new report.
Operations, facilities, support	EUR 360,000	Fixed partner support, incident handling, monitoring, facilities, and manual operations.
Total	EUR 1,300,000	Equals the assumed 2x competitor-normal baseline.

4. AWS Target Run-Rate

The AWS target is a fully loaded annual run-rate. It is not only the raw AWS service bill.

Direct AWS metered services, planning estimate: EUR 217,000
Platform operations, FinOps, support, run team: EUR 175,000
Security, governance, observability, DR reserve: EUR 65,000
Contingency and usage growth buffer: EUR 103,000
Target fully loaded AWS annual run-rate: EUR 560,000

The reason for this structure is that public AWS prices only price metered services. The project also needs production support, incident management, data governance, access reviews, cost control, DR drills, backups, tagging, audit evidence, and room for data volume variance.

4.1 Fully Loaded Target by Cost Area

AWS target cost area	Annual target	What is included
Compute and ETL	EUR 150,000	Glue, Data Quality, Kinesis, Firehose, Lambda, MWAA, batch/stream orchestration, and run support allocated to pipelines.
Storage and backup	EUR 95,000	S3 Raw/Curated/Serving zones, lifecycle transitions, backups, replication reserve, object monitoring, and restore testing.
BI and serving	EUR 135,000	Redshift, Athena, QuickSight, SPICE, dashboard serving, warehouse tuning, and performance support.
Operations, security, governance	EUR 180,000	CloudWatch, CloudTrail, KMS, Macie, Lake Formation administration, Terraform/CDK, support, FinOps, DR, and compliance work.
Total	EUR 560,000	Fully loaded run-rate used in the RFP response.

5. Direct AWS Metered Service Calculations

These calculations show the underlying service math. They are intentionally transparent so the team can defend the estimate and adjust volumes quickly.

5.1 Streaming Ingestion: Kinesis Data Streams

We use provisioned Kinesis for predictable POS/review streams because the store count is known and the workload can be capacity-planned. AWS example pricing shows a 4-shard stream costing USD 52.14 per month for shard hours plus PUT payload units.

Planning assumption:

3 production streams or equivalent stream capacity
Monthly cost per 4-shard stream = USD 52.14
Annual Kinesis cost = $52.14 * 3 * 12$ = USD 1,877

EUR equivalent = $1,877 * 0.92 = \text{EUR } 1,727$
Rounded with monitoring/retention buffer = EUR 2,500 / year

5.2 Stream Delivery: Amazon Data Firehose

Firehose lands streaming events into S3 and converts JSON to Parquet to reduce downstream Athena and Redshift Spectrum scan cost.

Planning assumption:

Monthly streaming data landed = 2 TB = 2,048 GB
Direct PUT / Kinesis-source ingestion = USD 0.029 per GB
Format conversion to Parquet = USD 0.018 per GB
Dynamic partitioning = USD 0.020 per GB

Monthly cost
= $2,048 * (0.029 + 0.018 + 0.020)$
= $2,048 * 0.067$
= USD 137.22

Annual cost
= $137.22 * 12$
= USD 1,646.64

EUR equivalent
= $1,646.64 * 0.92$
= EUR 1,515

Rounded with object/request buffer = EUR 2,000 / year

5.3 Batch ETL and Transformations: AWS Glue

Glue replaces the current cron-job ETL scripts. The model uses DPU-hours, not always-on servers.

Planning assumption:

Daily batch estate:
 $20 \text{ daily jobs} * 4 \text{ DPUs} * 1.5 \text{ hours/job} * 30 \text{ days} = 3,600 \text{ DPU-hours/month}$
Rounded planning usage = 3,500 DPU-hours/month

Glue ETL monthly cost
= $3,500 * \text{USD } 0.44$
= USD 1,540

Glue ETL annual cost
= $1,540 * 12$
= USD 18,480

Glue Data Quality:

DQ usage = 600 DPU-hours/month
DQ monthly cost = $600 * 0.44 = \text{USD } 264$
DQ annual cost = $264 * 12 = \text{USD } 3,168$

Crawlers/catalog/statistics/optimization:

Crawler and optimization reserve = 300 DPU-hours/month
Monthly cost = $300 * 0.44 = \text{USD } 132$
Annual cost = $132 * 12 = \text{USD } 1,584$

Data Catalog metadata:
Assume under first 1M objects/accesses during MVP = USD 0
Reserve for growth and partition churn = USD 100/month = USD 1,200/year

Total Glue planning cost:

Glue total USD = $18,480 + 3,168 + 1,584 + 1,200 = \text{USD } 24,432$
Glue total EUR = $24,432 * 0.92 = \text{EUR } 22,477$
Rounded = EUR 23,000 / year

5.4 Orchestration: Amazon MWAA

MWAA replaces cron with Airflow DAGs, dependency handling, retries, and operational visibility. AWS examples show a small environment around USD 449/month and a larger environment around USD 1,047/month.

Planning assumption:

```
Production environment plus non-production/test capacity = USD 1,500/month
Annual cost = 1,500 * 12 = USD 18,000
EUR equivalent = 18,000 * 0.92 = EUR 16,560
Rounded with log/storage buffer = EUR 18,000 / year
```

5.5 API Connectors and Event Tasks: AWS Lambda

Lambda handles scheduled pulls from external APIs, small enrichment tasks, and glue code around events. This is deliberately modeled with a high request count to show that Lambda remains small relative to the full platform.

Planning assumption:

```
Requests/month = 20,000,000
Memory = 512 MB = 0.5 GB
Average duration = 300 ms = 0.3 seconds
Free tier = 1,000,000 requests and 400,000 GB-seconds/month

Request charges:
Billable requests = 20,000,000 - 1,000,000 = 19,000,000
Monthly request cost = 19 * USD 0.20 = USD 3.80

Compute charges:
Monthly GB-seconds = 20,000,000 * 0.5 * 0.3 = 3,000,000 GB-seconds
Billable GB-seconds = 3,000,000 - 400,000 = 2,600,000
Monthly compute cost = 2,600,000 * 0.0000166667 = USD 43.33

Monthly Lambda cost = 3.80 + 43.33 = USD 47.13
Annual Lambda cost = 47.13 * 12 = USD 565.56
EUR equivalent = 565.56 * 0.92 = EUR 520
Rounded with API/logging buffer = EUR 1,000 / year
```

5.6 Storage and Lakehouse Retention: Amazon S3

The architecture keeps immutable raw data, curated Parquet, serving aggregates, and archives. Lifecycle policy moves cold data to cheaper classes.

Planning footprint:

Tier	Average footprint	Unit rate	Monthly cost
Raw hot data, S3 Standard	20 TB = 20,480 GB	USD 0.023 / GB-month	USD 471.04
Curated/Serving hot data, S3 Standard	30 TB = 30,720 GB	USD 0.023 / GB-month	USD 706.56
Warm history, S3 Standard-IA	60 TB = 61,440 GB	USD 0.0125 / GB-month	USD 768.00
Cold archive, S3 Glacier Flexible Retrieval	120 TB = 122,880 GB	USD 0.0036 / GB-month	USD 442.37
Storage subtotal	230 TB average	Mixed	USD 2,387.97 / month

Annual storage subtotal:

```
2,387.97 * 12 = USD 28,655.64
EUR equivalent = 28,655.64 * 0.92 = EUR 26,363
```

Lifecycle, request, replication, inventory, and backup testing reserve:

```
Reserve = 45% of storage subtotal
= 26,363 * 0.45
= EUR 11,863
```

```
Direct S3/lakehouse annual estimate
= 26,363 + 11,863
= EUR 38,226
Rounded = EUR 40,000 / year
```

The fully loaded storage/backup target is higher at EUR 95,000/year because it includes cross-region replication reserve, restore testing, object-lock/legal hold administration, privacy deletion workflows, and operational support.

5.7 Ad Hoc SQL on the Lakehouse: Amazon Athena

Athena is billed by TB scanned. Parquet, compression, and partition pruning are a major cost lever. AWS's own example shows a 3 TB text-file query costing USD 15, and the same analytical question dropping to USD 1.25 with compression plus Parquet column pruning.

Planning assumption:

```
Unoptimized scan volume = 60 TB/month
Reduction from Parquet + partition pruning = 70%
Optimized billed scan volume = 60 * (1 - 0.70) = 18 TB/month

Monthly Athena cost = 18 * USD 5 = USD 90
Annual Athena cost = 90 * 12 = USD 1,080
EUR equivalent = 1,080 * 0.92 = EUR 994
Ad hoc growth buffer = 3x
Rounded = EUR 3,000 / year
```

5.8 BI Warehouse: Amazon Redshift

Redshift serves curated BI workloads that need predictable concurrency and performance. The model uses a planning envelope rather than a single hard cluster shape because Redshift may be serverless or provisioned during implementation.

Planning assumption:

```
AWS public anchor:
Redshift Serverless begins at USD 1.50/hour.
Provisioned Redshift begins at USD 0.543/hour.
Managed storage example is USD 0.024/GB-month.

Base serverless/provisioned BI compute envelope = USD 3,500/month
Annual Redshift compute/storage estimate = 3,500 * 12 = USD 42,000
EUR equivalent = 42,000 * 0.92 = EUR 38,640
Rounded with snapshots/performance reserve = EUR 45,000 / year
```

This budget can be reduced with Serverless Reservations or Reserved Instances once actual usage patterns are known. AWS states Redshift Serverless Reservations can reduce compute costs by up to 45% for eligible workloads.

5.9 BI Dashboards: Amazon QuickSight

QuickSight replaces Excel distribution and manual reporting. The base model uses per-user pricing because Manga has a known internal audience.

Planning assumption:

```
Authors = 25
Readers = 400
SPICE total = 500 GB
Included SPICE = 25 authors * 10 GB = 250 GB
Billable SPICE = 500 - 250 = 250 GB

Author monthly cost = 25 * USD 24 = USD 600
Reader monthly cost = 400 * USD 3 = USD 1,200
SPICE monthly cost = 250 * USD 0.38 = USD 95

Monthly QuickSight cost = 600 + 1,200 + 95 = USD 1,895
Annual QuickSight cost = 1,895 * 12 = USD 22,740
EUR equivalent = 22,740 * 0.92 = EUR 20,921
```

Rounded = EUR 23,000 / year

5.10 Migration: AWS DMS

DMS is treated as a one-time migration/cutover cost, not a steady annual run-rate. The public AWS DMS page states that serverless/on-demand migration capacity is paid by the hour and scales with the data transaction volume. Because the case does not provide database size, exact DCUs, or cutover duration, we reserve a one-time implementation budget:

One-time DMS migration and CDC testing reserve = EUR 12,000
Included in implementation planning, not the recurring EUR 560,000 AWS run-rate

6. Direct AWS Subtotal

Service group	Annual direct estimate
Kinesis Data Streams	EUR 2,500
Data Firehose	EUR 2,000
AWS Glue ETL, Crawlers, Data Quality, Catalog	EUR 23,000
Amazon MWAA	EUR 18,000
AWS Lambda	EUR 1,000
Amazon S3 lakehouse direct storage/requests reserve	EUR 40,000
Amazon Athena	EUR 3,000
Amazon Redshift	EUR 45,000
Amazon QuickSight	EUR 23,000
CloudWatch, CloudTrail, KMS, Macie, Lake Formation, backup/monitoring reserve	EUR 60,000
Direct AWS metered-service subtotal	EUR 217,500

Rounded direct AWS service usage:

EUR 217,500 -> EUR 217,000

7. Why the Target Is EUR 560k, Not EUR 217k

The raw AWS services are only one part of the cost. A production enterprise data platform also requires people, support, compliance, change control, and contingency.

Direct AWS metered service estimate = EUR 217,000
Platform operations and FinOps = EUR 175,000
Security/governance/DR reserve = EUR 65,000
Subtotal before contingency = EUR 457,000

Contingency and usage growth buffer
= 457,000 * 22.5%
= EUR 102,825
Rounded = EUR 103,000

Fully loaded target
= 217,000 + 175,000 + 65,000 + 103,000
= EUR 560,000/year

This is the correct number to compare against current colocation because current colocation cost also includes support/operations overhead, not only hardware.

8. Savings Calculation

Current annual cost = EUR 1,300,000
AWS target annual cost = EUR 560,000

Annual savings
= 1,300,000 - 560,000
= EUR 740,000

Savings percentage
= 740,000 / 1,300,000
= 0.5692
= 56.9%
Rounded = 57%

9. Sensitivity Analysis

Scenario	Current baseline	AWS annual run-rate	Annual saving	Saving %	Interpretation
Optimistic	EUR 1,300,000	EUR 460,000	EUR 840,000	64.6%	Higher automation, lower data growth, strong Redshift/QuickSight optimization.
Base case	EUR 1,300,000	EUR 560,000	EUR 740,000	56.9%	Used in the proposal and cost figure.
Conservative	EUR 1,300,000	EUR 720,000	EUR 580,000	44.6%	Higher support needs, more data retention, more BI concurrency.

The base and conservative cases both sit inside the RFP's requested 40-60% reduction range.

10. Presentation Talking Points

Use this concise version on Slide 8:

Our cost model is not a raw AWS bill; it is a fully loaded annual run-rate. The RFP says Manga spends about 2x competitors, so we model the current state at EUR 1.30M/year. The AWS target is EUR 560k/year, built from public AWS unit prices plus operations, governance, DR, support, and contingency. That creates EUR 740k annual savings, or about 57%.

If asked for the details:

Glue is priced per DPU-hour, not per server.
Firehose and Kinesis are priced per GB/shard usage.
Athena is priced per TB scanned, which is why Parquet and partitioning are critical.
QuickSight is priced per Author/Reader plus SPICE.
Redshift is bounded with serverless/provisioned public price anchors.
S3 lifecycle policies reduce storage cost by moving cold history into IA and Glacier.

If asked what could change the estimate:

The largest variables are actual current invoices, real data volume, BI user count, required retention periods, support model, and final AWS Region. The formulas stay the same; we would replace assumptions with discovery numbers.

11. Data Needed to Finalize the Estimate

Before submitting a client-grade quote, collect:

Missing input	Why it matters
Current annual colocation and support invoices	Replaces the assumed EUR 1.30M baseline.
Exact AWS Region	AWS public prices vary by Region.
Monthly source data volume and growth rate	Drives S3, Firehose, Glue, Athena, Redshift, backup, and replication costs.

Missing input	Why it matters
Number of BI Authors, Readers, and embedded sessions	Drives QuickSight pricing model choice.
Required RTO/RPO and cross-region replication scope	Drives DR and backup reserve.
PII retention/legal-hold rules	Drives S3 lifecycle and deletion workflow cost.
Internal vs managed-service operations model	Drives the fully loaded run-rate above raw AWS service pricing.